Modeling Auckland Property Prices with Explainable AI

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***Abstract*—House prices affect lending, rates and major financial decisions, yet free online valuation tools give buyers a number without a reason. We build an explainable machine-learning model for Auckland house prices and test whether Auckland-specific external data (location, neighbourhood census and deprivation statistics, flood and aircraft-noise exposure, and development activity) improves prediction beyond a property's own characteristics. We link 59,708 Auckland sales (January 2024 to August 2026) to map coordinates and evaluate every model with five-fold, time-ordered cross-validation, so later sales never leak into earlier training folds. Adding 16 external features to a fixed XGBoost model lowers test RMSE from $343,718 to $212,678 and MAPE from 19.3% to 11.0%, a statistically significant gain, and a two-stage residual model shows that the external features explain 29.3% of the error a property-only model leaves. Permutation importance, SHAP and drop-group retraining agree that location carries most of the gain, while neighbourhood statistics largely repeat what location already says, and flood, noise and development features add almost nothing. XGBoost is the most accurate of five model families (MAPE 11.0%, against 15.7% for a hedonic regression), and SHAP turns each prediction into a short, per-sale explanation. A comparison with free online valuation estimates is still in progress.**

***Index Terms*—house price prediction, explainable AI, SHAP, gradient boosting, hedonic regression, Auckland**

I. Introduction

For most New Zealand households, a home is the largest thing they will ever buy, and its value affects many decisions. Buyers and sellers use valuations to set prices, banks use them to decide how much to lend, and Auckland Council uses them to set rates. In Auckland, these valuations are hard to get right. Homes vary widely in size, age and condition, and the region itself varies too: some areas are close to the city centre or the coast, some neighbourhoods are far wealthier than others, and some are exposed to flooding, aircraft noise or fast-changing development. A model that only looks at the house itself is therefore likely to miss a large part of its price.

Free online valuation tools give buyers a number but not a reason. A buyer cannot see whether an estimate is high because of the house, its location, or a flood risk the tool has picked up. This matters when the estimate is used to make a large financial decision.

Past research has looked at this problem from two directions. Hedonic studies of Auckland have measured the effect of single external factors on price, such as flood risk [1], school zones [2], leaky-building stigma [3] and social housing [4]. These models are easy to interpret, but each studies one factor at a time, and they are built to explain effects rather than to predict new sales [5]. Machine learning models predict prices more accurately than hedonic regression [6], [7], and SHAP [8] can show which features drive their predictions. However, we found no study that applies explainable machine learning to Auckland, and few that test how much a combined set of external features improves prediction for sales the model has not seen.

This paper addresses the following research questions:

- **RQ1:** Does adding Auckland-specific external features (location, neighbourhood, hazards and development) improve house price prediction compared with using only a property's own features?
- **RQ2:** How accurate and interpretable is our explainable model compared with a traditional hedonic model and free online valuation estimates for the same Auckland properties?

To answer these questions, we built a dataset of 59,708 single-home Auckland sales from January 2024 to August 2026. We linked each sale to a map coordinate using probabilistic record linkage, and used the coordinate to add 16 external features. We then trained the same XGBoost model twice, once with and once without the external features, using time-ordered cross-validation so that the model always learns from earlier sales and is tested on later ones. A paired bootstrap tests whether the difference is real, and a two-stage model tests whether external features explain errors that property features cannot. Three attribution methods show which groups of external features matter. We then compare XGBoost with random forest, CatBoost, a neural network and a hedonic regression on the same folds, and use SHAP to explain individual predictions.

Our main contributions are:

1) **A linked Auckland dataset.** We join council sales records to LINZ addresses using probabilistic record linkage, and add location, 2023 Census, deprivation, hazard and development features to each sale. We describe each cleaning step so that the dataset can be rebuilt.
2) **A fair test of external data.** By keeping the model and its settings fixed and training only on past sales, we isolate the effect of the external features. They reduce RMSE by 38.1% (from $343,718 to $212,678) and MAPE from 19.3% to 11.0%, a significant improvement, and on their own they explain 29.3% of the error left by a property-only model.

3) **Evidence on what drives the gain.** Location is the main source of the gain. Neighbourhood statistics carry real signal but overlap heavily with location, and hazard and development features add almost nothing at city scale.
4) **A comparison of model families with per-sale explanations.** XGBoost is the most accurate of five model families trained on identical inputs, with a MAPE of 11.0% against 15.7% for a hedonic regression. All five models agree on the most important features, and SHAP turns each prediction into a short explanation a buyer can read. [Online valuation benchmark (RQ2): pending.]

The rest of this paper is organised as follows. Section II reviews related work. Section III describes the data and how it was cleaned and linked. Section IV explains the experimental design. Section V presents and discusses the results. Section VI sets out the limitations, and Section VII concludes and suggests future work.

II. Related Work

A. Hedonic models and external factors in Auckland

Hedonic pricing treats a house as a bundle of characteristics, each with its own implicit price, which can be estimated by regressing sale price on those characteristics [9]. Hedonic models are mainly built to *explain* the effect of each factor, not to predict new sales as accurately as possible [5]. Auckland studies using this approach show that factors outside the property matter. Adding valuer-defined submarkets to a hedonic model improved prediction more than replacing it with more advanced spatial models [10]. Flood-prone homes on the North Shore sold for about 6.2% less before flood maps were published and 2.3% less afterwards [1]. Being in the zone of a popular school raised prices [2], while leaky-building stigma lowered house prices by about 5% [3], and nearby social housing lowered prices by 1.7–23% [4]. Studies outside New Zealand also show that ignoring the link between nearby sales can overstate the effect of an external factor such as flooding [11].

These studies have two limits. First, each one looks at a single external factor, so they do not show how much a combined set of external data improves prediction. Second, the models are linear, and the form of each effect (for example, how price changes with distance) must be chosen by hand. Our work instead adds 16 external features to one model at the same time and measures their combined effect on prediction error for sales the model has not seen.

B. Machine learning versus hedonic models

Tree-based machine learning models learn non-linear effects and interactions directly from the data. Across many studies they predict prices more accurately than hedonic regression. A random forest reached a test MAPE of about 7% against 17% for regression on Ljubljana apartments [6], and an R^2 of 0.94 against 0.65 on Busan condominiums [7]. In Russia, a random forest also beat regression (MAPE 14.9% vs 18.3%), although its error was much higher for larger apartments [12], so a single overall error can hide weak spots. Gradient-boosted trees such as XGBoost [13] perform similarly well. Soltani *et al.* [14] improved tree models further by adding a feature built from nearby recent sales, and stressed that such features must use only sales that happened *before* the sale being predicted.

Most of these comparisons change the model, the features, or both at once, so it is hard to tell how much of the gain comes from better data and how much from a better algorithm. We keep the algorithm and its settings fixed and change only the feature set, so any difference comes from the external data. When we compare algorithms, we do the opposite: every model gets the same features, folds and preprocessing. We also train only on past sales and test on later ones, which avoids leaking future information into the model.

C. External data and explainable models

Machine learning models are harder to interpret than hedonic models, which matters when buyers and lenders need to know why a home is valued as it is. SHAP [8] addresses this by splitting each prediction into the contribution of each feature. In Lisbon, adding open data on the surroundings reduced MAE by 8.24%, and SHAP showed that longitude and latitude were among the most important features [15]. In Sydney, SHAP showed that access to amenities affected price in a non-linear way that changed with distance and type of amenity [16]. In Beijing, location was the most important factor, and the effects of other features varied between areas [17]. SHAP, however, describes how the model uses each feature, not what causes prices to change, and it cannot show on its own that a model predicts well.

These studies use SHAP to rank features within one model. When features carry overlapping information, as coordinates and location-based features do, a single ranking can be misleading. We therefore first test whether external features improve prediction, and only then explain the model. To see how much each group of external features matters, we use three methods (permutation importance, SHAP and drop-group retraining) that react differently to overlapping features, and we confirm with a separate residual test that the external features add information the property features do not.

D. Summary

To our knowledge, no published study has applied explainable machine learning to Auckland house prices, and few have tested the joint predictive value of external features in a single model. Our work addresses both gaps. It also compares the model with four other model families, including a hedonic model, on identical folds.

III. Data

A. Data sources

Our main data is the Auckland Council sales file [18]. It lists 92,010 property sales from January 2024 to August 2026, with 41 columns that describe the sale (price, date, sale type) and the property (land area, floor area, building age, condition, garages, and so on).

We added external data from nine sources: Statistical Area 2 (SA2) boundaries [19] and 2023 Census totals for each SA2 [20]; the NZDep2023 deprivation index [21]; new dwellings consented in each SA2 from Stats NZ [22]; flood plain maps and the aircraft noise overlay from Auckland Council [23]; the New Zealand Schools Directory [24]; the LINZ coastline [25]; supermarket and road data from OpenStreetMap [26]; and bus stop locations from Auckland Transport [27].

B. Linking sales to coordinates

The sales file describes each property by its council valuation reference and a street address: a house number, an optional unit or letter, and a street name with an abbreviated road type (for example "St Michaels AV"). It has no coordinates and no suburb. To place each property on the map, we linked it to the LINZ NZ Addresses register [28], which gives a coordinate for every official address in New Zealand. The two sources share no key, and the same address is often written differently (abbreviations, unit formats, missing letters or typing errors), so an exact join on the address text would miss many properties and could not tell a near-miss from a wrong address.

We therefore used probabilistic record linkage with Splink [29], [30], an open-source implementation of the Fellegi–Sunter model [31]. For a candidate pair of records, the model compares several fields. Each comparison has a small number of ordered levels, such as exact agreement, close agreement under a string-similarity measure, disagreement, and missing. For level ℓ of comparison k, the model learns two probabilities: $m_{k\ell}$, the probability of seeing that level if the two records are the same address, and $u_{k\ell}$, the probability of seeing it if they are not. The evidence from all comparisons is combined into a match weight

$$w = \log_2 \frac{\lambda}{1-\lambda} + \sum_k \log_2 \frac{m_{k\gamma_k}}{u_{k\gamma_k}}, \qquad p = \frac{2^w}{1+2^w}, \tag{1}$$

where γ_k is the level observed for comparison k, λ is the prior probability that a random pair is a match, and p is the resulting match probability. Agreement on a field adds weight, and disagreement subtracts it. The more unusual the agreement is among non-matching pairs, the more weight it adds.

No hand-labelled pairs are needed to fit the model. The u probabilities are estimated from randomly sampled pairs, almost all of which are non-matches, and the m probabilities are estimated with the expectation–maximisation (EM) algorithm [29]. Splink can also adjust for term frequency, so that agreeing on a rare street name counts as stronger evidence than agreeing on a common one. To keep the number of pairs manageable, only pairs that pass a blocking rule are scored. [Blocking rules used, e.g. same street name, or same house number and first letters of the street name.] [Fields compared and their levels, e.g. house number (exact), unit/letter (exact or missing), street name (exact, Jaro–Winkler $\geq$ 0.9, else), road type; any cleaning applied first, such as lower-casing and expanding road-type abbreviations; whether term-frequency adjustment was used.]

Each valuation reference was assigned the LINZ address with the highest match weight, provided that the weight exceeded a threshold of [threshold, as a match weight or match probability]. [Clerical review: how many pairs were checked by hand, in which match-weight bands, and the false-match rate found.] The result links 81,363 of the 85,180 distinct properties in the sales file (95.5%) to a LINZ address and coordinate. Each property receives one address. A small number of LINZ addresses (72) are shared by two valuation references, as when two cross-leased homes share one street number. The coordinates are joined back to the sales by valuation reference, which combines the roll number, the assessment number and the lower-cased suffix that separates units sharing one assessment. After the filters in Table I, 59,798 of the 61,244 arm's-length single-home sales (97.6%) have a coordinate.

Once each property had a coordinate, we placed it in its 2025 SA2 to attach the census and deprivation values, and on the council maps to find whether it lies in a flood plain and which aircraft noise category applies. Distances to the city centre (Auckland CBD), the nearest open school and the coastline are straight-line distances in metres (NZTM). Walking distances to the nearest supermarket and bus stop were measured along the road network.

TABLE I
FILTERS APPLIED TO THE SALES DATA, IN ORDER

Filter	Sales left
All parsed sales	92,010
Residential, rural or lifestyle zoning	80,070
One unit of use	77,370
Freehold	76,395
One property per sale	74,112
Arm's-length market sale	61,244
Coordinate found	59,798
Exact duplicate records removed	59,711
Floor-area typos removed	59,708

C. Cleaning

We kept only sales that represent a normal market price for one home. Table I shows the filters, applied in order.

The zoning and unit filters remove commercial properties and buildings with several homes. The tenure filter removes leasehold sales, which sell for less because the buyer does not own the land. The one-property filter matters most: when several properties are sold together, the sale records one combined price but the features of only one property, so the model cannot learn from these rows. The arm's-length filter removes sales between related parties, which may not reflect market value. We then removed sales with no coordinate, records that repeat the same property, date and price, and three records whose building floor area is more than ten times the council's mass-appraisal floor area (for example 360,000 m^2 for an ordinary house), which are data-entry errors.

TABLE II
FEATURE SETS

Group	Features
Baseline (11)	
Size	land area, building floor area
Building	age, condition, construction type, property type, off-street parking, garages under main roof, freestanding garages
Time	sale year, sale month
External (16)	
Location (7)	latitude, longitude, distance to CBD, distance to coastline, distance to nearest school, walking distance to supermarket, walking distance to bus stop
Neighbourhood (6)	NZDep2023 index, median personal income, employment rate, unemployment rate, population, population density (all per SA2)
Hazards (2)	in flood plain, aircraft noise category
Development (1)	new dwellings consented in the SA2 (past 12 months)

We also removed five columns that would leak the answer to the model. The net, chattels and other price columns add up to the sale price. The council's capital value and land value are themselves price estimates, so a model that uses them would copy the council's valuation instead of learning from property and location data.

D. Features

We built two feature sets (Table II). The *baseline* set has 11 features that describe only the property and the sale date; it contains no information about location. The *augmented* set adds 16 external features, for 27 in total. All SA2 statistics are area-level values shared by every home in the same SA2.

Land area, floor area and the two walking distances are right-skewed, so we applied $\log(1+x)$ to them. Categories are converted to integer codes, and missing numbers (walking distances for 1.4–2.7% of sales) are filled with the median. Both steps are learned from the training data only (Section IV-B).

Removing near-duplicate features. Two features were dropped before modelling. Sale quarter is calculated from sale month, and the council's mass-appraisal floor area repeats the building floor area (Spearman $\rho = 0.96$); both pairs had variance inflation factors (VIF) of 17–18. Dropping them cost XGBoost $212 of RMSE (0.1%), and afterwards no feature has a VIF above 7.3. Removing them matters for interpretation more than accuracy: two near-copies of a feature split its SHAP credit between them and make both look less important than they are.

IV. METHODOLOGY

All code is available online.[1]

[1][https://github.com/ORG/REPO]. The dataset is built in `remake_dataset.ipynb`; the RQ1 experiments are in `remake_ablation.ipynb`; the model comparison, SHAP and per-sale explanations are in `remake_train.ipynb`, `remake_nn.ipynb` and `remake_traindition.ipynb`.

A. Models

Our main model is XGBoost [13], a gradient-boosted tree model that predicts the sale price in dollars. We chose it because it handles non-linear effects and mixed feature types well, and tree models have outperformed linear regression in earlier house price studies [6], [7]. It uses 500 trees, maximum depth 6, learning rate 0.05, 80% of rows and 80% of features sampled per tree, and random seed 42.

To check that our conclusions do not depend on one algorithm, we train four more models on exactly the same inputs:

- **Random forest:** 150 trees, maximum depth 18, at least 2 sales per leaf.
- **CatBoost:** 500 trees, depth 6, learning rate 0.05, set to match XGBoost. It receives the same integer-coded categories as the other models.
- **Neural network:** a multilayer perceptron with a learned embedding for each categorical feature, three hidden layers (256, 128 and 64 units, ReLU, dropout 0.1), trained with Adam (learning rate 10^{-3}, batch 512) on standardised log price. Training stops when the error on the latest 10% of the training period has not improved for 10 epochs. Inputs are clipped to the 0.5th–99.5th percentiles of the training data, and predictions are converted back to dollars with Duan's smearing factor (Section IV-G).
- **Hedonic regression:** see Section IV-G.

No model is tuned. If each model were tuned separately, part of any difference between them could come from tuning rather than from the features or the algorithm. Fixed settings keep the comparisons fair, at the cost of slightly understating what each model could reach.

B. Training and testing

House prices change over time, so a model should only learn from sales that happened before the sales it predicts. Randomly shuffling the data would let the model learn from future sales, which makes results look better than they would be in practice [14].

We therefore sort the 59,708 sales by date and use five-fold expanding-window cross-validation (Fig. 1). The data is cut into six consecutive time blocks of about 9,950 sales. In fold k, the model trains on blocks 1 to k and is tested on block $k+1$. Each fold therefore trains on more history than the one before. Every sale after the first block is tested exactly once, by a model that has never seen it. We do not keep a separate validation set because no settings are tuned.

Within each fold, all preprocessing is learned from the training blocks only: the category codes, the median values used to fill gaps, and the price limits described next. The test block is then processed with these training values.

Before training, we remove sales priced below the 1st percentile or above the 99th percentile of the training prices ($430,000–$450,000 and $4.2M–$4.5M, depending on the fold). The same limits are applied to the test block. This stops a few extreme sales from dominating the error. It also means

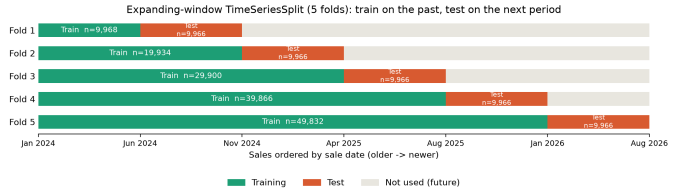


Fig. 1. Five-fold expanding-window cross-validation. Each fold trains on all earlier sales and tests on the next period.

our results describe the middle 98% of the market, not the very cheapest or most expensive homes. After this step, 48,777 test predictions remain across the five folds.

C. Evaluation metrics

For n test sales with actual price y_i and predicted price $\hat{y}_i$, we report three error measures:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{3}$$

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{y_i}. \tag{4}$$

MAE is the typical error in dollars. RMSE gives more weight to large errors, which matter most for expensive homes. MAPE is the error as a percentage of price, so a \$100K error on a \$600K home counts more than the same error on a \$3M home. We combine the test predictions from all five folds and compute each measure once on the combined set.

D. Testing whether the improvement is real

To check that the augmented model is better and not just lucky, we use a paired bootstrap. We draw n test sales at random with replacement and compute the difference

$$\Delta^{(b)} = M^{(b)}_{\text{base}} - M^{(b)}_{\text{aug}} \tag{5}$$

for each metric M. Both models are scored on the same resampled sales, so the comparison is fair. We repeat this $B = 2{,}000$ times. The 2.5th and 97.5th percentiles of $\Delta^{(b)}$ give a 95% confidence interval. If the interval does not include zero, the improvement is statistically significant.

E. Two-stage residual test

The test above shows whether external features help, but the augmented model could still use them to learn something the property features already contain. The two-stage test checks whether external features add *new* information.

In stage 1, we take the baseline model's out-of-fold test predictions and compute each sale's error $r_i = y_i - \hat{y}_i^{\text{base}}$. In stage 2, a new XGBoost model uses only the 16 external features to predict r_i. Stage 2 has no access to property features, so it can only succeed if location and the other external data explain what the property features missed.

Stage 2 uses its own five-fold expanding-window split over the 48,777 sales that have a stage-1 error, again sorted by date. The first block (8,132 sales) is used only for training, leaving 40,645 sales with a stage-2 prediction. We report the out-of-fold R^2 of stage 2 in Section V. A value above zero means the external features carry information the property features do not.

F. Which external features matter

For this analysis we use fold 5: the model trains on all sales up to January 2026 and is tested on the 9,769 sales from January to August 2026. We study the four external groups in Table II, and use three methods, because each one can be misleading on its own when features are correlated:

- **Grouped permutation importance.** We randomly shuffle all features of a group together in the test set and measure how much the RMSE increases. We repeat this 20 times and report the average.
- **SHAP.** We compute SHAP values [8] on 500 test sales. Each SHAP value is the amount, in dollars, that a feature moves one prediction up or down. We report each group's share of the total mean absolute SHAP value.
- **Drop-group retraining.** For each group (and each single external feature), we retrain the model without it and measure the change in test RMSE.

These methods react differently when two features carry similar information. Permutation importance breaks a group while the model still relies on it, so it shows how much the model *uses* the group, which can overstate how much it is *needed*. Drop-group retraining lets the model fall back on a similar feature in another group, which shows what is unique to the group but can understate its importance. SHAP splits the credit between similar features. Reading the three together shows both how much a group is used and how much of it is unique.

G. Hedonic baseline

To compare with the traditional approach, we fit a hedonic model: a linear regression on the logarithm of price,

$$\ln y_i = \beta_0 + \mathbf{x}_i^{\top}\boldsymbol{\beta} + \varepsilon_i, \tag{6}$$

estimated by ordinary least squares. It uses the same 27 features as the other models, so any difference in accuracy comes from the model, not the inputs. Categories are one-hot encoded with the most common level as the reference, and any column that is constant in a fold's training data is dropped for that fold. As for the neural network, numeric inputs are clipped to the training data's 0.5th–99.5th percentiles, so that a single unusual value cannot produce an extreme price through the exponential.

The model predicts log price, and turning this back into dollars with $\exp(\cdot)$ underestimates the average price. We correct this with Duan's smearing factor [32]:

$$\hat{y}_i = \exp\!\left(\hat{\beta}_0 + \mathbf{x}_i^{\top}\hat{\boldsymbol{\beta}}\right)\cdot\frac{1}{m}\sum_{k=1}^{m}e^{\hat{\varepsilon}_k}, \tag{7}$$

where $\hat{\varepsilon}_k$ are the m training residuals. The hedonic model uses the same folds, price limits and metrics as the other models, so all results are directly comparable.

H. Explaining individual predictions

SHAP values add up: for each sale, the model's average prediction plus the SHAP value of every feature equals the prediction. We use this to explain single predictions. For a held-out sale, we start from the average predicted price and add each feature's contribution in dollars. For a short summary, we sum the contributions within five themes (the four external groups plus "house and land"), which is valid because SHAP values are additive. A waterfall chart shows the largest individual contributions in real units (square metres, kilometres, deciles), so a reader can see which characteristics raised or lowered the price. For tree models we use exact TreeSHAP; for the neural network and hedonic model we use SHAP's permutation explainer with 100 training sales as background.

I. Online valuation benchmark

To test the second half of RQ2, we draw a sample of held-out, test-period sales and look up a free public valuation estimate for each one (e.g. homes.co.nz, propertyvalue.co.nz or trademe.co.nz), matched to the council record by address and confirmed against the actual sale date and price so we are comparing the estimate that was available *before* the sale, not one updated afterwards using the sale itself. We then compute the same RMSE/MAE/MAPE as the other models on this matched subset. [Sample size, match-confirmation results and the online-valuation error metrics.]

V. Experiments and Results

A. Does external data help?

Table III gives the pooled out-of-fold error over all five folds. Adding the 16 external features cuts RMSE by \$131,040, from \$343,718 to \$212,678 (a 38.1% relative reduction), MAE by \$95,242, and MAPE by 8.4 percentage points, from 19.3% to 11.0% (Fig. 2). The paired bootstrap (Eq. 5, $B = 2{,}000$) gives 95% confidence intervals of \$126,988–\$135,157 for the RMSE improvement, \$93,358–\$97,194 for MAE and 8.2–8.5 points for MAPE. None includes zero, so the gain is real and not a result of how the test sales happen to fall. The augmented model is also better in every one of the five folds, with RMSE between \$191,483 and \$238,373 against \$320,862 to \$370,069 for the baseline. In practical terms, an MAE of \$128,000 is still a large amount of money, but it is close to half the \$223,000 typical error of a model that only knows the house.

B. Do external features add new information?

A model could look like it needs external data only because the property features are weak, not because location genuinely carries new information. The two-stage test rules this out: a model using *only* the 16 external features explains 29.3% of the variance in what the property-only model gets wrong

TABLE III
Baseline (property only) vs. augmented (with external features), XGBoost, pooled out-of-fold over 5 folds, $n = 48{,}777$

Model	Features	RMSE	MAE	MAPE
Baseline	11	\$343,718	\$223,224	19.3%
Augmented	27	\$212,678	\$127,983	11.0%

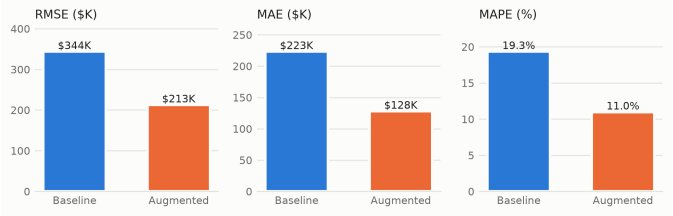


Fig. 2. Pooled out-of-fold error of the baseline and augmented XGBoost models.

($R^2 = 0.293$ on 40,645 held-out residuals), cutting the residual RMSE from \$339,582 to \$285,308. Since this second-stage model never sees a single property feature, this 29.3% must be information the property features do not already contain.

C. Which external features matter?

Table IV compares the three attribution methods from Section IV-F on fold 5 (9,769 test sales, augmented RMSE \$191,483). Together, the external features are 52.3% of the model's total mean absolute SHAP value, and dropping all of them raises RMSE by \$129,379, in line with the five-fold result above.

Location is the most important group on all three measures. Distance to the CBD alone is 15.5% of total SHAP, more than latitude (8.1%) and longitude (8.6%) individually, and second only to building floor area (21.0%) across all 27 features (Fig. 3).

The neighbourhood group shows why the three methods must be read together. The model clearly uses it: shuffling it raises RMSE by 32.1%, and it holds 16.1% of SHAP, led by the deprivation index (6.9%). Yet retraining without it costs only \$1,665, because the model can recover nearly the same information from location. The overlap runs both ways but is not symmetric: without the location group, the model still loses \$20,146 even though the neighbourhood statistics remain. Without all external features it loses \$129,379, more than six times the sum of the four single-group losses. In other words, location and neighbourhood largely describe the same thing (how desirable an area is), and location describes it more completely. The same pattern holds for single features: no external feature costs more than \$3,752 when dropped on its own (latitude), and for eight of them the model is marginally better without them, within noise.

Hazard and development features matter little at city scale. Only 6.4% of sales lie in a flood plain and 5.3% in an aircraft noise overlay, so these features can matter for those homes but cannot explain much of the variation in price across Auckland,

TABLE IV

ATTRIBUTION OF THE EXTERNAL FEATURE GROUPS ON FOLD 5. PERMUTATION: RMSE INCREASE WHEN THE GROUP IS SHUFFLED. SHAP: SHARE OF TOTAL MEAN |SHAP|. DROP-GROUP: RMSE CHANGE WHEN THE MODEL IS RETRAINED WITHOUT THE GROUP (NEGATIVE = MARGINALLY BETTER WITHOUT IT).

Group	Permutation	SHAP	Drop-group
All external (16)	+171.2%	52.3%	+$129,379
Location (7)	+131.1%	35.1%	+$20,146
Neighbourhood (6)	+32.1%	16.1%	+$1,665
Hazards (2)	+0.3%	0.4%	+$370
Development (1)	+0.6%	0.6%	−$1,489

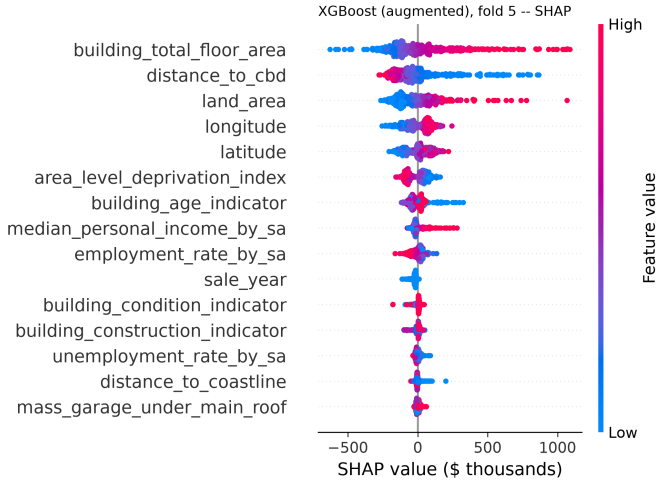


Fig. 3. SHAP summary for the augmented XGBoost model on 500 fold-5 test sales (15 most important features). Each dot is one sale; colour is the feature's value.

and the 12-month count of consents in an SA2 changes slowly and overlaps with location.

D. Comparison of model families

Table V compares the five models on the same 48,777 test sales and the same 27 features. XGBoost is the most accurate on every metric (MAPE 11.0%), followed closely by CatBoost and random forest (11.6%), then the neural network (12.3%). The hedonic regression is clearly the least accurate (15.7%). All four machine-learning models train all five folds in under a minute on a CPU.

The models also agree on what matters. Table VI shows the SHAP share of the leading features in each model, computed on the same 500 fold-5 test sales. Building floor area is first and distance to the CBD second in all five models, and the coordinates are among the top features in every model. The main difference is that the hedonic model leans more on the SA2 income and employment statistics: a linear model cannot learn sharp local patterns from latitude and longitude, so it relies on smoother area-level summaries instead.

E. Comparison with a hedonic model (RQ2)

The hedonic regression has the same inputs as XGBoost but is 4.7 MAPE points less accurate (15.7% vs 11.0%), and its RMSE is 44% higher ($306,564 vs $212,678). It explains 72.9% of the variance in log price on fold 5's training data, so it is a reasonable model, but a straight line in each feature cannot capture the non-linear and interacting effects a tree model learns directly, such as a distance to the CBD that matters a great deal near the city and very little far from it (Fig. 3).

In exchange, each hedonic coefficient has a direct, one-sentence interpretation that holds for every house. On fold 5, for example, each extra $10,000 of median personal income in the SA2 is associated with 12.3% higher prices, each kilometre further from the CBD with 1.2% lower prices, each step up the deprivation index with 1.3% lower prices, and a flood plain location with 1.8% lower prices, all else equal. SHAP gives a comparable dollar breakdown for the more accurate model, but a feature's effect can change from one sale to another. This is the accuracy/interpretability trade-off RQ2 asks about: the machine-learning model keeps a per-sale explanation (Section V-F) while cutting the hedonic model's MAPE by about 30%.

TABLE V

ALL MODELS ON THE SAME FEATURES AND FOLDS, POOLED OUT-OF-FOLD, $n = 48{,}777$. TIME IS FOR TRAINING ALL FIVE FOLDS.

Model	RMSE	MAE	MAPE	Time
XGBoost	$212,678	$127,983	10.98%	4 s
CatBoost	$220,406	$135,087	11.61%	10 s
Random forest	$228,130	$134,876	11.55%	14 s
Neural network	$247,526	$148,230	12.29%	49 s
Hedonic (OLS, log)	$306,564	$186,708	15.72%	4 s

TABLE VI

SHARE OF TOTAL MEAN |SHAP| (%) FOR LEADING FEATURES, SAME 500 FOLD-5 TEST SALES. RF = RANDOM FOREST, NN = NEURAL NETWORK.

Feature	XGB	CatB	RF	NN	Hedonic
Building floor area	21.0	21.8	31.6	16.8	21.0
Distance to CBD	15.5	14.6	18.9	16.6	11.0
Land area	12.7	14.0	13.6	11.5	3.8
Latitude + longitude	16.7	17.2	19.1	13.6	11.2
Deprivation index	6.9	8.1	3.7	7.0	3.4
Median income (SA2)	3.4	3.9	3.3	4.5	8.6
Building age	4.7	4.5	2.1	4.7	8.5

F. Explaining a single prediction

Fig. 4 shows the explanation for one held-out sale in fold 5's test period, a home sold in June 2026 for $1,285,000. The XGBoost model, trained only on sales up to January 2026, predicted $1,266,131 (1.5% below). Starting from the average prediction of $1,151,033, the neighbourhood raises the price by $65,399, mostly through a low deprivation index (2 of 10, +$71,055); location adds $41,119, with the property's latitude (+$85,033) outweighing its 16.9 km distance from the CBD (−$63,763); the house and land add $26,184, as a large floor area (172 m^2, +$84,080) more than offsets a small section (186 m^2, −$94,139); and the 2026 market timing subtracts $19,070. Hazards change the price by only $1,469. This is

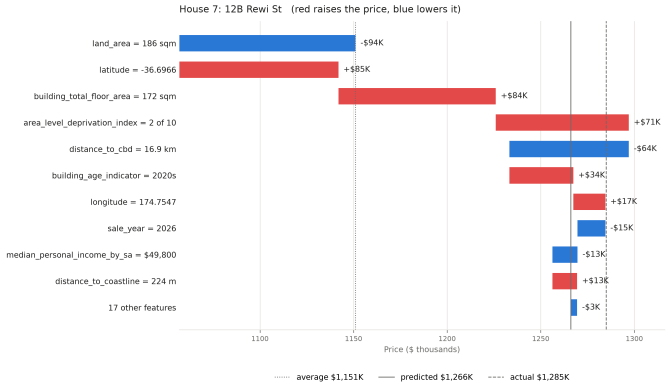


Fig. 4. SHAP explanation of one held-out sale (XGBoost, fold 5). Bars run from the average prediction to the model's prediction; red raises the price and blue lowers it. Lines mark the average, predicted and actual price.

the kind of reason an online valuation tool does not give: the buyer can see that the price rests on the area and the size of the house, not on the section.

G. Comparison with online valuation estimates (RQ2)

[Online valuation benchmark results: see Section IV-I – sample size, matched properties, and RMSE/MAE/MAPE on the matched subset, alongside the augmented model's error on the same sales.]

VI. Limitations

Our price filter keeps only the middle 98% of each fold's training prices (Section IV-B), so our results describe the typical Auckland home and may not transfer to the cheapest or most expensive 1% of sales. Record linkage could not place a coordinate for every sale: 61,244 arm's-length single-property sales were available, and 59,798 (97.6%) were matched to a coordinate (Section III-B); if the unmatched 2.4% are systematically different (e.g. new subdivisions or non-standard addresses not yet in LINZ), our results could be slightly optimistic or pessimistic for that slice of the market, in a direction we have not separately quantified.

The neighbourhood features are 2023 Census and NZDep2023 values for the whole SA2, so every home in an SA2 shares them, and they do not change over the 2024–2026 sales period. They describe the area, not the household that buys the home. Our attribution methods measure association with the model's predictions, not causation: a high SHAP value for the deprivation index says the model leans on that feature, not that deprivation *causes* a specific dollar change in price. Because location and neighbourhood features overlap, the split of credit between them (Section V-C) depends on the method; we report three methods for this reason, but no single number should be read as the "true" value of one feature. Finally, we deliberately fixed every model's hyperparameters for a fair comparison; a project optimising purely for accuracy could tune each model and would likely do somewhat better than every number reported here.

VII. Conclusion and Future Work

This paper asked whether Auckland-specific external data improves house price prediction (RQ1), and how an explainable model built on it compares with a traditional hedonic model and free online valuations (RQ2). For RQ1, the answer is yes, and by a wide and statistically significant margin: adding 16 external features cuts XGBoost's RMSE by 38.1% and MAPE by 8.4 percentage points, and a two-stage test confirms that the external features explain 29.3% of the error a property-only model leaves. For the first half of RQ2, XGBoost is the most accurate of five model families and far more accurate than a hedonic regression on the same features (MAPE 11.0% vs 15.7%), while SHAP still explains each prediction in dollars; the second half, comparison against free online valuation estimates, is still being completed (Section IV-I).

The clearest practical finding is that most of the value of external data in this project is the value of *location*. Distance to the CBD and the coordinates carry the largest share; area-level census and deprivation statistics are useful but mostly repeat what location already says; and flood, noise and development features add almost nothing to a city-wide model. For a team building a similar model with a limited data budget, accurate geocoding and a few well-chosen location measures are worth more than many hand-engineered area features.

Future work should (1) finish and report the online valuation benchmark; (2) tune each model and check whether the ranking of model families holds; (3) test whether hazard features matter more in the parts of Auckland where they are common, for example with location–feature interaction methods such as GeoShapley [33], rather than in the city-wide average reported here; and (4) extend the record-linkage pipeline to recover the 2.4% of sales that currently have no matched coordinate, to check whether they change any of these conclusions.

Contributions

[Member [C.] 1]: [Contribution.]
[Member [C.] 2]: [Contribution.]
[Member [C.] 3]: [Contribution.]
[Member [C.] 4]: [Contribution.]
[Member [C.] 5]: [Contribution.]

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